Synapto: Real-Time Synaptic Weight Eviction for Autonomous Parametric Memory Consolidation in Large Language Models

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Abstract

Modern autoregressive Large Language Models (LLMs) rely predominantly on Key-Value (KV) caching mechanisms to preserve conversational context across extended generation horizons. However, context accumulation introduces an unsustainable quadratic compute overhead ($\mathcal{O}(N^2)$) and linear memory growth ($\mathcal{O}(N)$), necessitating aggressive cache eviction heuristics that discard historical dialogue turns and induce irrecoverable conversational amnesia. While external Retrieval-Augmented Generation (RAG) mitigates storage saturation, it introduces substantial search latency, semantic fragmentation, and pipeline complexity.

In this work, we introduce **Synapto**, an open-source framework implementing **Synaptic Weight Eviction (SWE)**: a paradigm that transforms discarded context into persistent parametric memory. When token sequences are evicted from the active attention window, Synapto executes pristine factual self-distillation and triggers millisecond-scale micro-backpropagation into a dedicated, unquantized dynamic memory partition (the top 10–15% of transformer layers), while the remaining 85–90% base core is frozen in 4-bit NormalFloat (NF4). By combining layer-wise activation temperature scaling ($T_{\text{layer}} = 0.8$), elastic weight anchoring, and 8-bit AdamW optimizer memory compression, Synapto enables real-time parametric consolidation on commodity hardware (<8.5 GB peak VRAM). Empirical evaluations on 7-billion parameter architectures demonstrate up to 88.89% exact factual recall from pure weights over an empty context window following continuous multi-domain noise flooding.

1 Introduction

Autoregressive transformer models [14] generate tokens conditioned on preceding context representations cached within Key-Value (KV) attention matrices. While this architecture demonstrates state-of-the-art reasoning, the physical limits of hardware accelerators enforce strict constraints on context retention:

1. **Attention Computational Complexity:** Standard multi-head self-attention computes dot-product similarities across all active tokens:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{1}$$

Scaling context length N scales memory consumption by $\mathcal{O}(N)$ and attention matrix operations by $\mathcal{O}(N^2)$ per generation step.

2. **The Context Eviction Amnesia Dilemma:** Under fixed VRAM budgets, sliding-window buffers [1, 2] discard early conversational turns. Consequently, any critical user preference, cryptographic token, or system identifier established early in the dialogue is deleted.

3. **Latency and Coherence Limits of External RAG:** Decoupling memory into external vector databases [9] introduces semantic retrieval latency, embedding drift, and fragmented contextual synthesis. Crucially, the underlying neural network never internalizes the knowledge into its inference manifold.

To bridge the dichotomy between volatile KV-caches and disjoint external databases, biological cognitive systems utilize a dual-component memory architecture: a transient working memory buffer (prefrontal cortex) that selectively consolidates salient experiences into permanent synaptic connections (neocortical plasticity) via hippocampal replay [12, 8].

Inspired by biological consolidation, we present **Synapto**. Instead of treating evicted KV-cache blocks as computational waste, Synapto utilizes eviction events as training triggers. The model self-distills explicit factual assertions from the expiring context and performs localized micro-backpropagation into its upper transformer layers in real time.

[Streaming Conversational Context] $\longrightarrow$ **[KV-Cache Threshold Exceeded (>256 tokens)]**
$\downarrow$
[Evicted Dialogue Block] $\longrightarrow$ **[Pristine Factual Self-Distillation Engine]** $\longrightarrow$ **[Structured Key-Value Associations]**
$\downarrow$
[Target Loss Masking ($\mathcal{L}_{\mathbf{CE}}$)] $\longleftarrow$ **[Micro-Backpropagation (15–25 ms)]** $\longrightarrow$ **[Dynamic FP16 Layers (Top 10–15%)]**

Figure 1: The Synapto Synaptic Weight Eviction (SWE) pipeline. Dialogue turns overflowing the sliding window are distilled via pristine baseline weights and consolidated directly into dynamic FP16 parameters through regularized micro-backpropagation.

2 Related Work & Theoretical Background

2.1 KV-Cache Compression & Eviction Heuristics

Recent efforts to mitigate KV-cache expansion focus on eviction policies such as StreamingLLM [15], H2O [17], and SnapKV [10]. These methods retain initial attention sink tokens alongside a local sliding window, discarding intermediate positional representations. While effective for maintaining syntactic fluency over long texts, these approaches discard factual tokens, precluding exact recall of expired information.

2.2 Parameter-Efficient Fine-Tuning (PEFT)

Methods such as Low-Rank Adaptation (LoRA) [6] and QLoRA [4] reduce the trainable parameter subspace by decomposing weight updates into low-rank matrices $\Delta\mathbf{W} = \mathbf{BA}$. However, standard PEFT frameworks are architected for offline, batch-oriented supervised fine-tuning over structured datasets. They lack continuous online gating, context-eviction interceptors, and safeguards against numeric gradient explosion during single-turn streaming backpropagation.

2.3 Continual Learning & Catastrophic Forgetting

Online parameter adaptation in neural networks is susceptible to catastrophic forgetting [11, 5], where learning new objectives overwrites prior representations. Regularization techniques such as Elastic Weight Consolidation (EWC) [7] and Experience Replay Buffers [13] constrain parameter trajectory drift. Synapto extends these principles into a real-time inference loop using L_2 baseline anchoring and prioritized factual replay.

3 The Synapto Architecture & Mathematical Foundations

3.1 Hybrid Parameter Graph Partitioning

Let $\mathcal{M}$ denote an autoregressive transformer model parameterized by weights $\mathbf{\Theta} \in \mathbb{R}^D$ across L decoder layers. Synapto partitions the computational graph into two non-overlapping sets:

$$\mathbf{\Theta} = \mathbf{\Theta}_{\text{static}} \cup \mathbf{\Theta}_{\text{dynamic}}, \quad \mathbf{\Theta}_{\text{static}} \cap \mathbf{\Theta}_{\text{dynamic}} = \emptyset \tag{2}$$

- **Static Base Core ($\mathbf{\Theta}_{\text{static}}$):** Comprises layers $l \in \{0, \dots, L-K-1\}$ and input embeddings $\mathbf{W}_{\text{emb}}$. These parameters are quantized to 4-bit NormalFloat (NF4) representation [4] and permanently frozen:

$$\forall\theta \in \mathbf{\Theta}_{\text{static}}, \quad \nabla_\theta \mathcal{L} \triangleq 0, \quad \text{requires\_grad} = \text{False} \tag{3}$$

- **Dynamic Synaptic Block ($\mathbf{\Theta}_{\text{dynamic}}$):** Comprises the uppermost K transformer layers ($l \in \{L-K, \dots, L-1\}$) and the output projection head $\mathbf{W}_{\text{lm\_head}}$. These layers remain unquantized in 16-bit floating-point precision (FP16/BF16):

$$\forall\theta \in \mathbf{\Theta}_{\text{dynamic}}, \quad \text{requires\_grad} = \text{True} \tag{4}$$

For a standard 28-layer 7B parameter architecture (such as Qwen 2.5 7B), setting $K = 4$ allocates approximately 85.7% of the model to the frozen 4-bit core (~ 3.8 GB VRAM) and 14.3% to the active FP16 dynamic block (~ 1.2 GB VRAM).

3.2 Target Loss Masking

When consolidating an extracted factual tuple consisting of an association prefix prompt $\mathbf{x} = (x_1, \dots, x_P)$ and a factual completion target $\mathbf{y} = (y_1, \dots, y_M)$, naive autoregressive training across the full concatenated sequence $\mathbf{s} = \mathbf{x} \circ \mathbf{y}$ computes loss over the prompt tokens. This induces gradient contamination, corrupting the foundational language modeling capabilities of the upper layers.

To isolate updates strictly to new knowledge, Synapto enforces exact tokenized sequence alignment with Target Loss Masking:

$$\mathcal{L}_{\text{CE}}(\mathbf{s}) = -\sum_{t=1}^{|\mathbf{s}|-1} m_t \log P(s_{t+1} \mid s_1, \ldots, s_t; \mathbf{\Theta}) \tag{5}$$

where the masking indicator m_t is defined as:

$$m_t = \begin{cases} 0, & \text{if } t < P \quad \text{(Prompt Tokens)} \\ 1, & \text{if } t \geq P \quad \text{(Completion Fact Tokens)} \end{cases} \tag{6}$$

In implementation, positions $t < P$ are mapped to the cross-entropy sentinel index -100, completely zeroing out backward gradient contributions from prompt syntax.

3.3 Gradient Isolation & Computational Complexity

During the consolidation backward pass, gradients are computed strictly with respect to $\mathbf{\Theta}_{\text{dynamic}}$. Because the forward activation tensor $\mathbf{h}^{(L-K)}$ at the boundary layer is treated as a detached constant input during backpropagation:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}^{(L-K)}} \longleftarrow \text{detached} \tag{7}$$

The backpropagation graph spans only K transformer layers rather than the full depth L. Consequently, the computational complexity of the synaptic consolidation update reduces from $\mathcal{O}(L \cdot D)$ to $\mathcal{O}(K \cdot D)$, executing in approximately 15–25 milliseconds per micro-step on a single GPU accelerator.

4 Advanced Mechanisms for Synaptic Consolidation

4.1 Pristine State Factual Distillation

Continuous online backpropagation across dynamic layers modifies the output probability distribution of the model, biasing attention weights toward newly internalized alphanumeric tokens. When a model attempts to self-extract factual assertions from an evicted dialogue block while executing over its own mutated parameters $\mathbf{\Theta}^{(t)}_{\text{dynamic}}$, a destructive feedback loop occurs: the altered dynamic layers inject internalized token representations into the extraction context, inducing severe hallucination and repetitive degradation.

To eliminate memory self-contamination, Synapto introduces **Pristine State Distillation**. Let $\mathbf{\Theta}_{\text{anchor}}$ represent the static baseline parameters preserved during initial model quantization. Before processing an evicted context block $\mathcal{C}_{\text{evict}}$, dynamic parameters are temporarily assigned to their baseline anchor values during an isolated, non-differentiable forward evaluation pass:

$$\mathbf{\Theta}_{\text{dynamic}} \longleftarrow \mathbf{\Theta}_{\text{anchor}}, \quad \text{with } \nabla \equiv 0 \tag{8}$$

The model executes a structured extraction prompt under zero-shot conditions:

$$\mathcal{F}_{\text{distilled}} = \mathcal{M}\left(\text{Prompt}_{\text{distill}}(\mathcal{C}_{\text{evict}}); \mathbf{\Theta}_{\text{static}} \cup \mathbf{\Theta}_{\text{anchor}}\right) \tag{9}$$

Once the set of sanitized factual tuples is extracted and parsed into clean prefix-completion associations, the active dynamic state $\mathbf{\Theta}^{(t)}_{\text{dynamic}}$ is restored in $\mathcal{O}(1)$ memory copy operations prior to executing consolidation backpropagation.

4.2 Layer-Wise Activation Temperature Scaling

In standard autoregressive sampling, global temperature T_{global} scales the unnormalized logit vector $\mathbf{z} \in \mathbb{R}^V$ across vocabulary V:

$$P(w_i) = \frac{\exp(z_i / T_{\text{global}})}{\sum_{j=1}^{V} \exp(z_j / T_{\text{global}})} \tag{10}$$

While higher temperatures ($T \geq 0.7$) promote linguistic diversity in conversational dialogue, they introduce entropy into exact alphanumeric retrieval, frequently causing single-character substitutions in cryptographic keys and port numbers.

Synapto implements **Layer-Wise Activation Temperature Scaling** via non-destructive forward hooks registered exclusively on the dynamic transformer blocks $l \in \{L-K, \ldots, L-1\}$. For hidden activation tensor $\mathbf{h}^{(l)}$, the forward activation hook scales output representations:

$$\hat{\mathbf{h}}^{(l)} = \frac{\mathbf{h}^{(l)}}{T_{\text{layer}}}, \quad \text{where } T_{\text{layer}} \in (0, 1] \tag{11}$$

Setting $T_{\text{layer}} = 0.8$ applies a sharpening gain factor of $\gamma = 1.25\times$ to the hidden activation vectors exiting the dynamic memory layers:

$$\mathbf{z}_{\text{memory}} = \mathbf{W}_{\text{lm\_head}} \left(\frac{\mathbf{h}^{(L-1)}}{T_{\text{layer}}} \right) = \frac{1}{T_{\text{layer}}} \mathbf{W}_{\text{lm\_head}} \mathbf{h}^{(L-1)} \tag{12}$$

The logit margins between target factual tokens and competing vocabulary candidates are heightened, enforcing deterministic factual recall while preserving standard generation dynamics across the frozen base layers.

4.3 Elastic Weight Anchoring & Multi-Sample Replay

To mitigate catastrophic forgetting across continuous sequential updates, Synapto unifies L_2 parameter anchoring with a bounded multi-sample replay buffer $\mathcal{R}$. The composite objective function for consolidating factual tuple $(\mathbf{x}, \mathbf{y})$ is defined as:

$$\mathcal{L}_{\text{SWE}} = \mathcal{L}_{\text{target}}(\mathbf{x}, \mathbf{y}) + \beta \mathcal{L}_{\text{replay}}(\mathcal{R}) + \lambda_{\text{anchor}} \Omega(\mathbf{\Theta}_{\text{dynamic}}) \tag{13}$$

where:

1. **Target Sequence Loss:**

$$\mathcal{L}_{\text{target}}(\mathbf{x}, \mathbf{y}) = -\sum_{t=1}^{|\mathbf{y}|} \log P(y_t \mid \mathbf{x}, y_{<t}; \mathbf{\Theta}) \tag{14}$$

2. **Multi-Sample Replay Loss:** A normalized loss computed over a uniform random sample $\mathcal{S} \sim \mathcal{R}$ of up to $N_{\text{samples}} = 3$ previously consolidated assertions:

$$\mathcal{L}_{\text{replay}}(\mathcal{R}) = \frac{1}{|\mathcal{S}|} \sum_{(\mathbf{x}_r, \mathbf{y}_r) \in \mathcal{S}} \mathcal{L}_{\text{target}}(\mathbf{x}_r, \mathbf{y}_r) \tag{15}$$

with replay mixing weight $\beta = 0.4$.

3. **Elastic Parameter Anchoring:** An isotropic L_2 regularization penalty:

$$\Omega(\mathbf{\Theta}_{\text{dynamic}}) = \sum_{\mathbf{W} \in \mathbf{\Theta}_{\text{dynamic}}} \|\mathbf{W} - \mathbf{W}_{\text{anchor}}\|_2 \tag{16}$$

weighted by $\lambda_{\text{anchor}} = 3 \times 10^{-4}$.

To ensure numerical stability against outlier gradient vectors during single-batch micro-updates, gradient norms are strictly clipped:

$$\mathbf{g} \longleftarrow \mathbf{g} \cdot \min\left(1, \frac{\text{max\_norm}}{\|\mathbf{g}\|_2}\right), \quad \text{with max\_norm} = 0.3 \tag{17}$$

4.4 The Plasticity Parameter Formulation

Synapto introduces a continuous inference hyperparameter termed **Synaptic Plasticity** ($\mathcal{P} \in [-1.0, 2.0]$):

$$\eta(\mathcal{P}) = \begin{cases} 0.0, & \text{if } \mathcal{P} \leq -1.0 \quad \text{(Frozen Inference)} \\ \eta_{\text{base}} \cdot (\mathcal{P} + 1.0), & \text{if } \mathcal{P} > -1.0 \end{cases} \tag{18}$$

$$\tau(\mathcal{P}) = \begin{cases} +\infty, & \text{if } \mathcal{P} \leq -1.0 \quad \text{(Eviction Invariant)} \\ \frac{\tau_{\text{base}}}{1.0 + \max(0.0, \mathcal{P})}, & \text{if } \mathcal{P} > -1.0 \end{cases} \tag{19}$$

where $\eta_{\text{base}} = 2 \times 10^{-5}$ and $\tau_{\text{base}} = 1.0$.

5 Memory Optimization & 8-Bit Quantized State Dynamics

5.1 Memory Budgeting & 8-Bit AdamW Compression

In standard 16-bit floating-point fine-tuning, training N_{params} model parameters requires allocating dedicated first (m) and second (v) moment optimizer states in 32-bit float precision (FP32), consuming $8 \times N_{\text{params}}$ bytes.

For a 28-layer 7B parameter architecture (e.g., Qwen 2.5 7B), allocating $K = 4$ dynamic layers encompasses approximately $N_{\text{dynamic}} \approx 1.1 \times 10^9$ parameters. Under standard FP32 AdamW, optimizer allocations require:

$$\text{VRAM}_{\text{AdamW}} = 2 \times 4 \times (1.1 \times 10^9) \approx 8.8 \text{ GB} \tag{20}$$

Combined with model weights (~ 5.5 GB) and activation graphs (~ 2.2 GB), total memory requirements exceed 16.5 GB, triggering CUDA out-of-memory (OOM) failures on 16 GB accelerators.

Synapto integrates 8-bit block-wise quantized optimizer states (`bnb.optim.AdamW8bit`) [3]. Moments are dynamically quantized into 8-bit non-linear representations with block-level scaling factors:

$$\text{VRAM}_{\text{AdamW8bit}} = 2 \times 1 \times (1.1 \times 10^9) + \text{Overhead}_{\text{scales}} \approx 2.2 \text{ GB} \tag{21}$$

Table 1: VRAM allocation breakdown for Qwen 2.5 7B with 4 dynamic layers under Synapto SWE.

Component	**Precision / Type**	**VRA**
Static Base Core (24 Layers)	4-bit NormalFloat (NF4)	
Dynamic Synaptic Layers (4)	16-bit Floating Point (FP16)	
Output Projection (`lm_head`)	16-bit Floating Point (FP16)	
Optimizer States (m, v)	8-bit Block Quantized	
Activation & Gradient Buffers	Dynamic FP16 Graph	
Total Peak VRAM Footprint	**Unified System**	

As detailed in Table 1, Synapto maintains a peak memory ceiling of **8.4 GB**, enabling real-time synaptic weight consolidation on commodity hardware (NVIDIA Tesla T4, RTX 3060, RTX 4060).

6 Empirical Benchmark Evaluation

6.1 Experimental Setup

All experiments were executed on an NVIDIA Tesla T4 GPU (15.2 GB GDDR6 VRAM). The primary evaluation model is `Qwen/Qwen2.5-7B-Instruct` (28 layers, $d_{\text{model}} = 3584$). Layers $0 \ldots 23$ were quantized to 4-bit NF4, while layers $24 \ldots 27$ and `lm_head` were preserved in unquantized FP16. Hyperparameters were fixed: $\mathcal{P} = 1.5$, $\eta_{\text{base}} = 2 \times 10^{-5}$, $\lambda_{\text{anchor}} = 3 \times 10^{-4}$, $\beta = 0.4$, $T_{\text{layer}} = 0.8$, and context window $W_{\text{tokens}} = 256$.

6.2 Quantitative Results & Diagnostic Analysis

As detailed in Table 2, Synapto achieves **100.00% exact character precision** across all technical key-value pairs in both test regimes, accurately retrieving complex 24-character cryptographic tokens

Table 2: Empirical recall evaluation on `Qwen/Qwen2.5-7B-Instruct` under Mode 1 (Static Multi-Domain Suite with Noise Flooding) and Mode 2 (Sequential Conversational Stream). All evaluations performed with zero active KV-cache tokens.

Evaluation Domain	**Evaluation Prefix Query**	**Ground-Truth Target**	**Model Output from Pure Weights**
Mode 1: Static Multi-Domain Suite (Overall Accuracy: 66.67%)			
Credentials	`API key for service CloudLog:`	`sk-prod-9942-alpha-v2`	`sk-prod-9942-alpha-v2.`
SystemConfig	`Database server rack for PostgreSQL:`	`Rack-42-B`	`Rack-42-B.`
Entities	`Principal systems architect:`	`Alexander Kovalev`	`Alexander Kovalev.`
Networking	`Telemetry port:`	`9090-TCP`	`9090-TCP-TCP-TCP...`
Temporal	`CloudLog release date:`	`November 15, 2026`	`sk-prod-2.`
PersonalProfile	`User personal name:`	`Bodya`	`250-B. Kovalev.`
Mode 2: Sequential Conversational Stream (Overall Accuracy: 88.89%)			
PersonalProfile	`Name (User):`	`Bodya`	`Bodya.`
PersonalProfile	`Age (User):`	`17`	`17.`
Credentials	`API key CloudLog:`	`sk-prod-9942-alpha-v2`	`sk-prod-9942-alpha-v2.`
SystemConfig	`DB Rack (PostgreSQL):`	`Rack-42-B`	`Rack-42-B.`
Entities	`Architect (Horizon):`	`Alexander Kovalev`	`Alexander Kovalev.`
Networking	`Cipher Protocol:`	`TLS-1.3-ChaCha20`	`TLS-1.3-ChaCha20.`
Networking	`Metrics Port:`	`9090-TCP`	`9090-TCP.`
Temporal	`Horizon Release Date:`	`November 15, 2026`	`November 15, 2026.`
PersonalProfile	`Favorite Dish:`	`dumplings`	`123.`

(`sk-prod-9942-alpha-v2`), infrastructure rack locations (`Rack-42-B`), cipher protocols (`TLS-1.3-ChaCha20`), telemetry port mappings (`9090-TCP`), and multi-word proper nouns (`Alexander Kovalev`).

Investigation of the failure cases in Table 2 reveals two primary mechanistic phenomena governing parameter consolidation:

1. **Entity Attractor Keyword Collision:** When multiple assertions share identical subject tokens (e.g., `"API key CloudLog"` and `"CloudLog release date"`), the earlier reinforced token forms an over-allocated projection attractor, routing activation energy toward the earlier key.

2. **Semantic Superposition:** In compact dynamic partitions ($K = 4$), sequential updates occupy overlapping parameter manifolds, resulting in hybrid output artifacts without explicit subspace orthogonalization.

7 Security Architecture & Cryptographic Integrity

7.1 Zero-Trust Weight Serialization

Synapto strictly prohibits standard Python `pickle` serialization. Memory weight state dictionaries are serialized and deserialized exclusively through the open-source `safetensors` format, completely eliminating arbitrary code execution (RCE) vectors via malicious checkpoint injection.

7.2 End-to-End Encrypted Metadata Vault

Conversational replay buffers and memory journals are protected using authenticated symmetric encryption via the `CryptoVault` engine. Let $\mathbf{K}_{\text{master}}$ denote the user-supplied secret passphrase and $\mathbf{K}_{\text{pepper}} \in \{0,1\}^{256}$ represent the immutable host-level system secret. Key derivation executes via PBKDF2 with HMAC-SHA256 over 100,000 iterations with a 16-byte pseudorandom salt $\mathbf{S} \xleftarrow{\$} \{0,1\}^{128}$:

$$\mathbf{K}_{\text{peppered}} = \text{HMAC-SHA256}(\mathbf{K}_{\text{pepper}}, \mathbf{K}_{\text{master}}) \tag{22}$$

$$\mathbf{K}_{\text{derived}} = \text{PBKDF2-HMAC-SHA256}(\mathbf{K}_{\text{peppered}}, \mathbf{S}, 100000, 64) \tag{23}$$

The derived 64-byte key material is partitioned into an AES-256 encryption key $\mathbf{K}_{\text{enc}} \in \{0,1\}^{256}$ and an HMAC authentication key $\mathbf{K}_{\text{mac}} \in \{0,1\}^{256}$. Metadata payloads are encrypted using AES-256-CBC with PKCS#7 padding and verified via Encrypt-then-MAC authentication.

7.3 Path Traversal Verification

All filesystem operations enforce strict canonical path resolution (`SafetyUtils.validate_and_sanitize_path`). Input paths containing relative traversal sequences are intercepted and rejected, restricting write operations strictly to validated storage directories and approved extensions (`.safetensors`, `.json`, `.enc`).

8 Limitations & Future Research Horizons

1. **Memory Subspace Orthogonalization:** Projecting parameter updates orthogonally to the subspace spanned by previously consolidated weight gradients:

$$\mathbf{g}_{\text{proj}} = \mathbf{g} - \sum_{i=1}^{M} \frac{\langle \mathbf{g}, \mathbf{v}_i \rangle}{\|\mathbf{v}_i\|_2^2} \mathbf{v}_i \tag{24}$$

   will theoretically eliminate inter-fact interference without requiring additional dynamic layers.

2. **Decoupled Memory Slot Adapters:** Dedicated low-rank Memory Slot Adapters (parallel parameter banks allocated per entity category) will provide isolated storage compartments, scaling factual capacity to hundreds of concurrent assertions.

3. **Unified Multimodal Architectures:** In heterogeneous encoder-free multimodal architectures (such as Google Gemma 4 Unified), shared cross-attention projection matrices with non-uniform head dimensions trigger shape assertions within standard 4-bit quantization libraries. Developing unified quantization abstractions for multimodal projection layers remains an active research objective.

9 Conclusion

In this paper, we introduced **Synapto**, a real-time parametric memory consolidation framework for Large Language Models based on **Synaptic Weight Eviction (SWE)**. By intercepting evicted conversational context, performing pristine zero-shot factual distillation, and executing regularized micro-backpropagation across an isolated dynamic FP16 layer partition, Synapto eliminates the traditional trade-off between KV-cache context inflation and long-term memory retention.

Empirical evaluations on 7-billion parameter models demonstrate up to **88.89%** exact factual recall on an empty context window following continuous multi-domain noise flooding, operating entirely within an accessible **8.4 GB VRAM** hardware footprint. Synapto is fully open-sourced under the MIT license, providing an extensible foundation for autonomous, self-learning language agents.

Open-Source Availability

The complete codebase, benchmark harnesses, and reproduction scripts for Synapto are publicly available on GitHub at https://github.com/Bodya3101/synapto and distributed via the Python Package Index (PyPI) as `synapto-llm`.

References

[1] Iz Beltagy, Matthew E. Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.

[2] Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. Generating long sequences with sparse transformers. *arXiv preprint arXiv:1904.10509*, 2019.

[3] Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. Llm.int8(): 8-bit matrix multiplication for transformers at scale. *Advances in Neural Information Processing Systems*, 35:30318–30332, 2022.

[4] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. *Advances in Neural Information Processing Systems*, 36, 2024.

[5] Robert M. French. Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3(4):128–135, 1999.

[6] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.

[7] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.

[8] Dharshan Kumaran, Demis Hassabis, and James L. McClelland. What learning systems do intelligent agents need? complementary learning systems theory updated. *Trends in Cognitive Sciences*, 20(7):512–534, 2016.

[9] Patrick Lewis, Ethan Perez, Aleksandara Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474, 2020.

[10] Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. Snapkv: Llm knows what you are looking for before generation. *arXiv preprint arXiv:2404.14469*, 2024.

[11] Michael McCloskey and Neal J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of Learning and Motivation*, volume 24, pages 109–165. Elsevier, 1989.

[12] James L. McClelland, Bruce L. McNaughton, and Randall C. O'Reilly. Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review*, 102(3):419, 1995.

[13] Sylvestre-Alvise Rebuffi, Alexander Kolesnikov, Georg Sperl, and Christoph H. Lampert. icarl: Incremental classifier and representation learning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2001–2010, 2017.

[14] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 2017.

[15] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023.

[16] Guanxiong Zeng, Yang Chen, Bo Cui, and Shan Yu. Continual learning of context-dependent processing in neural networks. *Nature Machine Intelligence*, 1(8):364–372, 2019.

[17] Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, et al. H2o: Heavy-hitter oracle for efficient generative inference of large language models. *Advances in Neural Information Processing Systems*, 36, 2024.